STEP 1 — Install and Import Required Libraries

```
!pip install torch transformers datasets peft accelerate
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: peft in /usr/local/lib/python3.12/dist-packages (0.18.1)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.13.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.7
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.2
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from peft) (5.9.5)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->t
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformer
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->data
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspe
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->hugg
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datase
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hugg
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hu
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0
```

```
import torch
from transformers import AutoModelForSequenceClassification, AutoTokenizer
from datasets import load_dataset
from peft import LoraConfig, get_peft_model
```

STEP 2 — Load Pre-trained Model and Tokenizer

```
model_name = "distilbert-base-uncased"

# Load tokenizer
tokenizer = AutoTokenizer.from_pretrained(model_name)

# Load pre-trained model
model = AutoModelForSequenceClassification.from_pretrained(
```

```
    model_name,
    num_labels=2
)

print("Model loaded successfully")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%  483/483 [00:00<00:00, 5.78kB/s]
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 1.27kB/s]
vocab.txt: 232k/? [00:00<00:00, 3.10MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
tokenizer.json: 466k/? [00:00<00:00, 15.6MB/s]
model.safetensors: 100%  268M/268M [00:02<00:00, 252MB/s]
Loading weights: 100%  100/100 [00:00<00:00, 400.21it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
pre_classifier.bias      | MISSING    |
pre_classifier.weight    | MISSING    |
classifier.weight        | MISSING    |
classifier.bias          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Model loaded successfully
```

STEP 3 — Load and Preprocess Dataset

```
dataset = load_dataset("imdb")
```

```
README.md: 7.81k/? [00:00<00:00, 126kB/s]
plain_text/train-00000-of-00001.parquet: 100%  21.0M/21.0M [00:00<00:00, 105MB/s]
plain_text/test-00000-of-00001.parquet: 100%  20.5M/20.5M [00:00<00:00, 102MB/s]
plain_text/unsupervised-00000-of-00001.p(…): 100%  42.0M/42.0M [00:00<00:00, 108MB/s]
Generating train split: 100%  25000/25000 [00:00<00:00, 64397.12 examples/s]
Generating test split: 100%  25000/25000 [00:00<00:00, 58577.59 examples/s]
Generating unsupervised split: 100%  50000/50000 [00:00<00:00, 100660.82 examples/s]
```

```
def tokenize_function(example):
    return tokenizer(example["text"], truncation=True, padding="max_length")
```

```
tokenized_dataset = dataset.map(tokenize_function, batched=True)
```

```
Map: 100%  25000/25000 [00:34<00:00, 720.41 examples/s]
Map: 100%  25000/25000 [00:34<00:00, 737.13 examples/s]
Map: 100%  50000/50000 [01:08<00:00, 725.92 examples/s]
```

```
train_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(500))
test_dataset = tokenized_dataset["test"].shuffle(seed=42).select(range(200))
```

```
print("Dataset ready")
```

```
Dataset ready
```

STEP 4 — Configure LoRA Adapter

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)
```

STEP 5 — Apply PEFT Model

```
peft_model = get_peft_model(model, lora_config)
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
```

```
peft_model.print_trainable_parameters()
```

```
trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925
```

STEP 6 — Train the PEFT Model

```
from transformers import TrainingArguments, Trainer
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)
```

```
trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset
)
```

```
trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

[63/63 24:30, Epoch 1/1]

Step	Training Loss
10	0.716769
20	0.671785
30	0.671734
40	0.650887
50	0.651515
60	0.633901

```
TrainOutput(global_step=63, training_loss=0.6640450557072958, metrics={'train_runtime': 1504.7058, 'train_samples_per_second':
0.332, 'train_steps_per_second': 0.042, 'total_flos': 67369703424000.0, 'train_loss': 0.6640450557072958, 'epoch': 1.0})
```

STEP 7 — Evaluate and Generate Predictions

```
from transformers import TrainingArguments, Trainer

training_args = TrainingArguments(
```

```
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)

trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset
)
```

```
trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

[63/63 24:47, Epoch 1/1]

Step	Training Loss
10	0.593720
20	0.549042
30	0.553187
40	0.436252
50	0.423107
60	0.367281

```
TrainOutput(global_step=63, training_loss=0.4839775600130596, metrics={'train_runtime': 1519.3739, 'train_samples_per_second':
0.329, 'train_steps_per_second': 0.041, 'total_flos': 67369703424000.0, 'train_loss': 0.4839775600130596, 'epoch': 1.0})
```

```
results = trainer.evaluate()

print(results)
```

[25/25 03:18]

```
{'eval_loss': 0.4094918966293335, 'eval_runtime': 208.0362, 'eval_samples_per_second': 0.961, 'eval_steps_per_second': 0.12, 'ep
```

STEP 8 — Compare with Full Fine-Tuning

```
from transformers import TrainingArguments, Trainer

training_args_full = TrainingArguments(
    output_dir="./full_finetune",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)

trainer_full = Trainer(
    model=model,
    args=training_args_full,
    train_dataset=train_dataset,
    eval_dataset=test_dataset
)

trainer_full.train()
```

[63/63 23:44, Epoch 1/1]

Step	Training Loss
10	0.296066
20	0.333662
30	0.386744
40	0.303780
50	0.341355
60	0.334772

```
TrainOutput(global_step=63, training_loss=0.3363017581758045, metrics={'train_runtime': 1455.8341, 'train_samples_per_second':
0.343, 'train_steps_per_second': 0.043, 'total_flos': 67369703424000.0, 'train_loss': 0.3363017581758045, 'epoch': 1.0})
```

```
results_full = trainer_full.evaluate()

print("Full Fine-Tuning Evaluation:", results_full)
```

[25/25 03:28]

```
Full Fine-Tuning Evaluation: {'eval_loss': 0.38252896070480347, 'eval_runtime': 216.1124, 'eval_samples_per_second': 0.925, 'eva
```

```
total_params = sum(p.numel() for p in model.parameters())

trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)

print("Total Parameters:", total_params)
print("Trainable Parameters:", trainable_params)
```

```
Total Parameters: 67694596
Trainable Parameters: 739586
```

```
peft_model.print_trainable_parameters()
```

```
trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925
```